



Chapter - 8

Electromagnetic Waves



1. INTRODUCTION

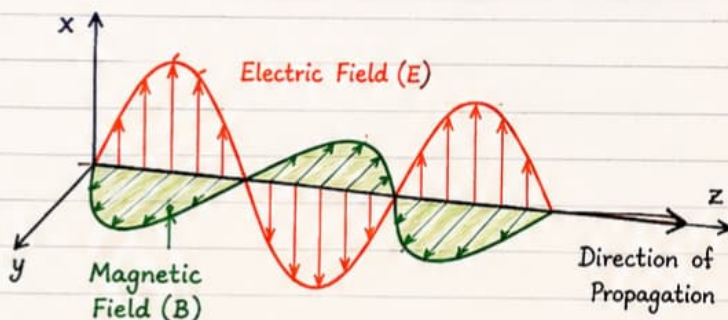
- Electromagnetic waves (EM waves) are transverse waves consisting of oscillating **electric** and **magnetic** fields that propagate through space, even in **vacuum**, without the need of a material medium.
- These waves are produced due to accelerating charges.
- They carry **energy** and **momentum**.

Key Points

- ★ Produced by accelerating charges.
- ★ Do not require any medium.
- ★ Transverse in nature.
- ★ Travel with speed of light in vacuum ($c = 3 \times 10^8$ m/s).
- ★ Obey laws of reflection, refraction, interference, diffraction & polarization.

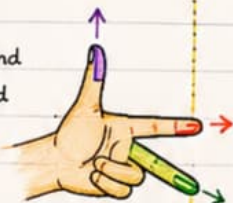
2. NATURE OF ELECTROMAGNETIC WAVES

- EM waves are **transverse**.
- Electric field (**E**) and Magnetic field (**B**) are:
 - Perpendicular to each other
 - Perpendicular to the direction of propagation.



Right Hand Thumb Rule

Stretch the thumb, forefinger and middle finger of your right hand mutually perpendicular to each other.



- Forefinger** → Electric field (**E**)
- Middle finger** → Magnetic field (**B**)
- Thumb** → Direction of propagation

Important Relation

In free space,

$$c = \frac{E_0}{B_0}$$

Where,

- E_0 = Amplitude of electric field
- B_0 = Amplitude of magnetic field
- c = Speed of light in vacuum (3×10^8 m/s)

3. PROPERTIES OF EM WAVES

- Electromagnetic waves are transverse.
- They do not require any material medium.
- They travel in vacuum with speed $c = 3 \times 10^8$ m/s.
- They can be reflected, refracted, diffracted, interfered and polarized.
- They carry energy and momentum.
- Obey inverse square law.
- Produced by accelerating charges.



4. WAVE EQUATIONS

- Speed of EM wave,
- Wavelength (λ), frequency (ν) and speed are related as,
- Electric and magnetic fields at a point are related as,

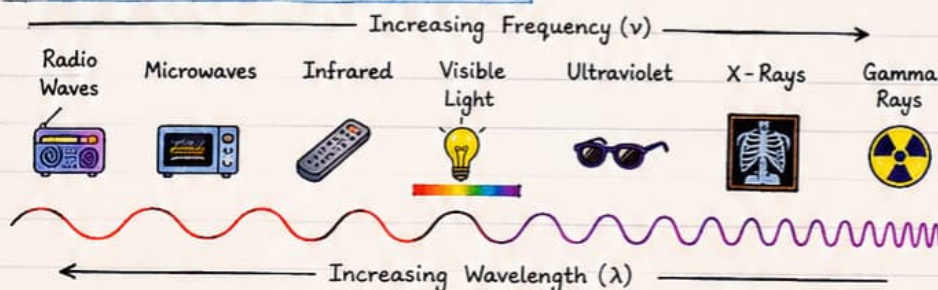
$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$c = \nu \lambda$$

$$E_0 = c B_0$$

Where, μ_0 = Permeability of free space
 ϵ_0 = Permittivity of free space

5. SPECTRUM OF ELECTROMAGNETIC WAVES



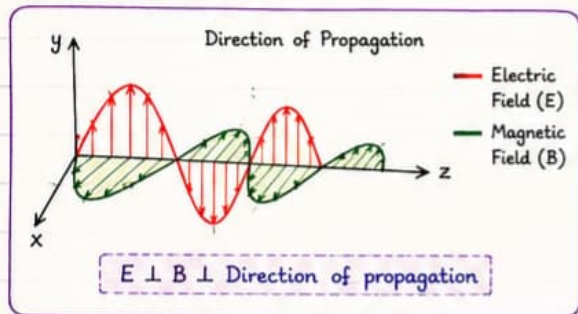
Note

Visible light is only a small part of the entire EM spectrum.

All EM waves travel with the same speed in vacuum but have different wavelengths and frequencies.

6. PROPAGATION OF EM WAVES

- EM waves **do not need any medium**.
- They can travel in **vacuum**.
- In free space, the magnitude of electric field and magnetic field at all points are perpendicular to each other and to the direction of propagation.
- EM waves are produced by **accelerating charges**. The changing electric field produces magnetic field and changing magnetic field produces electric field.



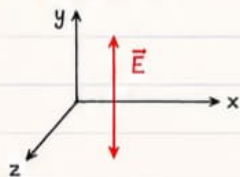
8. POLARIZATION OF EM WAVES

- Polarization is the phenomenon of restricting the electric field vector of a transverse wave in a particular direction.
- Only transverse waves can be polarized.

Types of Polarization

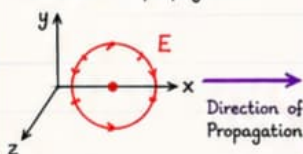
Linear Polarization

Electric field oscillates in one fixed direction.



Plane (or) Circular Polarization

Electric field rotates in a plane perpendicular to the direction of propagation.



Uses: Polaroid glasses, 3D movies, photography, LCD screens.

10. PRODUCTION OF EM WAVES

- Produced by accelerating charges.
- Common sources:
 - ★ Hertz's experiment (oscillating charges)
 - ★ Radio antennas
 - ★ Microwave ovens (magnetron)
 - ★ X-ray tubes
 - ★ Atoms and molecules (during electronic transitions)



7. ENERGY AND MOMENTUM

- EM waves carry energy and momentum.

Energy Density (u)

Energy per unit volume stored in EM wave,

$$u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2 \quad (\text{J/m}^3)$$

In free space,
 $u_E = u_B = u/2$

Poynting Vector (S)

It represents the rate of flow of energy per unit area in the direction of propagation.

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) \quad (\text{W/m}^2)$$

Intensity (I)

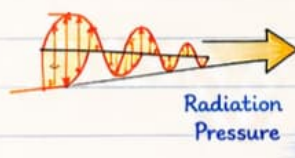
Average power per unit area is called intensity.

$$I = \langle S \rangle = \frac{c \epsilon_0 E_0^2}{2} = \frac{c B_0^2}{2\mu_0} \quad (\text{W/m}^2)$$

EM waves exert pressure,

$$P = \frac{I}{c}$$

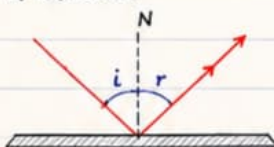
(Radiation Pressure)



9. REFLECTION, REFRACTION & OTHER PHENOMENA

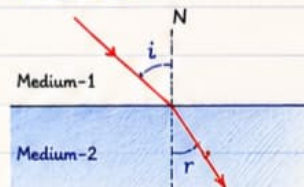
Reflection

EM waves obey the laws of reflection.



Refraction

When EM wave enters from one medium to another, its speed and wavelength change but frequency remains same.



Diffraction

Bending of EM waves around obstacles or through narrow openings.



Interference

Superposition of waves results in redistribution of intensity.



EM waves obey all the general properties of waves. ☆

11. ELECTROMAGNETIC SPECTRUM - DETAILS

Radiation	Radio Waves	Microwaves	Infrared (IR)	Visible Light	Ultraviolet (UV)	X-Rays	Gamma Rays
Wavelength (λ) (m)	$> 10^{-1}$	$10^{-1} - 10^{-3}$	$10^{-3} - 7 \times 10^{-7}$	$7 \times 10^{-7} - 4 \times 10^{-7}$	$4 \times 10^{-7} - 10^{-8}$	$10^{-8} - 10^{-11}$	$< 10^{-11}$
Frequency (ν) (Hz)	$< 3 \times 10^9$	$3 \times 10^9 - 3 \times 10^{11}$	$3 \times 10^{11} - 4.3 \times 10^{14}$	$4.3 \times 10^{14} - 7.5 \times 10^{14}$	$7.5 \times 10^{14} - 3 \times 10^{16}$	$3 \times 10^{16} - 3 \times 10^{19}$	$> 3 \times 10^{19}$
Energy ($E = h\nu$)	Very low	Low	Moderate	Moderate (visible)	High	Very high	Extremely high
Sources	Radio transmitters, TV towers	Radar, ovens, satellite	Hot bodies, heat lamps	 Sun, bulbs, LED	Sun, arc lamps	X-ray tubes	Nuclear reactions
Uses / Effects	Communication, broadcasting	Cooking, radar, Wi-Fi, satellite	Thermal imaging, remote controls	Vision	Sterilization, fluorescence, cause sunburn	Medical imaging, security check, crystallography	Cancer treatment, sterilization

Important Note

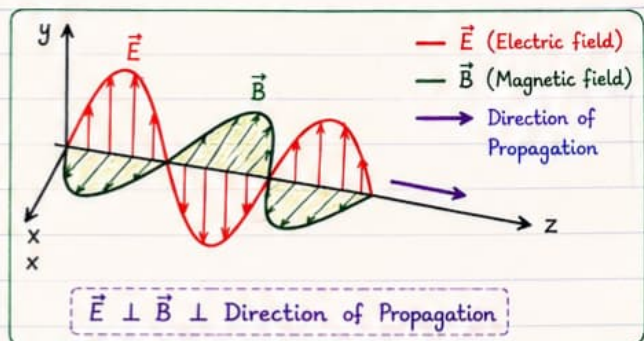
In vacuum all EM waves travel with same speed $c = 3 \times 10^8 \text{ m/s}$ but have different wavelengths and frequencies. ☆



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12. EM WAVES FROM ACCELERATING CHARGES

- Whenever a charge **accelerates**, it emits EM waves.
- A time-varying **electric field** produces a time-varying **magnetic field** and vice-versa.
- These changing fields detach from the source and propagate through space.

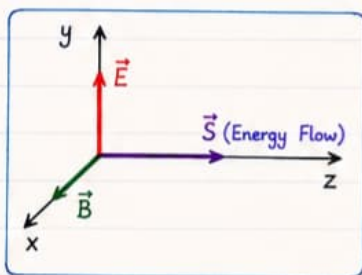


13. Poynting Vector - Energy Flow in EM Waves

The Poynting vector represents the direction of energy flow.

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

Unit : W/m^2



Properties of $\vec{S}$:

- $\vec{S}$ is perpendicular to both $\vec{E}$ and $\vec{B}$.
- Direction of $\vec{S}$ is the direction in which energy is transmitted.
- Magnitude of $\vec{S}$ is intensity I .

15. PRESSURE EXERTED BY EM WAVES (RADIATION PRESSURE)

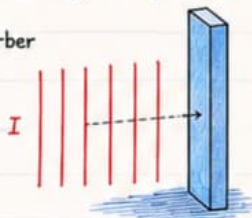
When EM waves are incident on a surface, they exert pressure.

$$P = \frac{I}{c}$$

For perfect absorber

For perfectly reflecting surface

$$P = \frac{2I}{c}$$



14. INTENSITY OF EM WAVES

Average power per unit area normal to the direction of propagation.

$$I = \frac{1}{\mu_0} E_{rms} B_{rms}$$

Also,

$$I = \frac{1}{2} c \epsilon_0 E_0^2$$

or

$$I = \frac{1}{2} \frac{c}{\mu_0} B_0^2$$

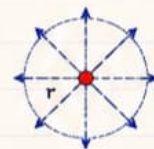
$$E_{rms} = \frac{E_0}{\sqrt{2}}$$

$$B_{rms} = \frac{B_0}{\sqrt{2}}$$

Variation of Intensity

For a source emitting power P uniformly in all directions,

$$I = \frac{P}{4\pi r^2}$$



17. SUMMARY OF KEY RELATIONS

Quantity	Expression / Relation	Quantity	Expression / Relation
① Speed of EM wave in vacuum	$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$	⑥ Intensity (in terms of E_0)	$I = \frac{1}{2} c \epsilon_0 E_0^2$
② Relation between E and B	$E_0 = c B_0$	⑦ Intensity (in terms of B_0)	$I = \frac{1}{2} \frac{c}{\mu_0} B_0^2$
③ Wave equation relation	$v = \lambda \nu$	⑧ Radiation pressure (absorber)	$P = \frac{I}{c}$
④ Poynting vector	$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$	⑨ Radiation pressure (reflector)	$P = \frac{2I}{c}$
⑤ Intensity	$I = \frac{1}{\mu_0} E_{rms} B_{rms}$	⑩ Momentum per unit area	$\frac{I}{c}$

18. REAL LIFE APPLICATIONS OF EM WAVES

Radio Waves	Microwaves	Infrared	Visible Light	Ultraviolet	X-Rays	Gamma Rays
Broadcasting, communication	Radar, Wi-Fi, Satellite communication	Thermal imaging, Heat sensing, TV remote	Vision, bulbs, Photography, Photosynthesis	Sterilization, Fluorescent lamps, Vitamin D synthesis	Medical imaging, Security check, Crystallography	Cancer treatment, Nuclear reactions, Sterilization

Quick Recall

- EM waves are transverse.
- $\vec{E}$, $\vec{B}$ and direction are mutually perpendicular.
- They need no medium.
- They carry energy and momentum.





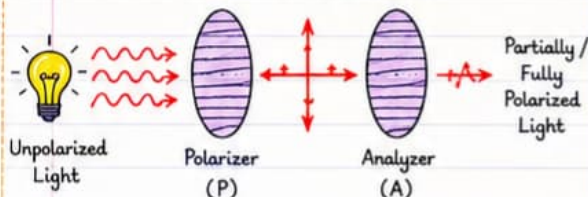
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19. POLARIZATION IN PRACTICE

- Unpolarized light has electric field in all directions perpendicular to the direction of propagation.
- Polarizing filters allow only one direction of vibration to pass.
- Polaroid sheets, 3D glasses, LCD screens use polarization.

Use of Polaroid Filters



Intensity varies as: $I = I_0 \cos^2 \theta$
(Malus' Law)

θ = Angle between polarizer and analyzer axes

20. EM WAVES AND SPEED OF LIGHT

- All EM waves travel with the same speed in vacuum
 $c = 3 \times 10^8 \text{ m/s}$.

- In a medium, speed $v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$

- Refractive index (n) of a medium,

$$n = \frac{c}{v} = \sqrt{\epsilon_r \mu_r}$$

In free space,
 $\epsilon_r = 1, \mu_r = 1$
 $\Rightarrow v = c$

21. EXAMPLE PROBLEMS (SHORT)

- Q1. Calculate the wavelength of a radio wave of frequency $1.5 \times 10^7 \text{ Hz}$.

Sol. $\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{1.5 \times 10^7} = 20 \text{ m}$

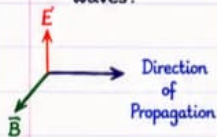
- Q2. The electric and magnetic field amplitudes in an EM wave are 6.0 N/C and $2.0 \times 10^{-8} \text{ T}$ respectively. Find the speed of the wave.

Sol. $c = \frac{E_0}{B_0} = \frac{6.0}{2.0 \times 10^{-8}} = 3 \times 10^8 \text{ m/s}$

22. SUMMARY - PROPERTIES OF ELECTROMAGNETIC WAVES

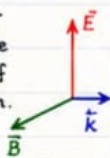
1

Transverse waves.



2

E and B are perpendicular to each other and to the direction of propagation.



3

Do not require any material medium.



Travel in vacuum.

4

Speed in vacuum
 $c = 3 \times 10^8 \text{ m/s}$



5

Carry energy and momentum.



6

Can be reflected, refracted, diffracted, interfered and polarized.



7

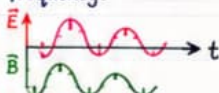
Obey inverse square law.



$$I \propto \frac{1}{r^2}$$

8

Electric and magnetic fields are in phase and have same frequency.



9

Spectrum ranges from radio waves to gamma rays.



10

Source: Accelerating charges.



23. QUICK COMPARISON

Property	EM Waves	Mechanical Waves
Need of Medium	Not required	Required
Nature	Transverse	Longitudinal or Transverse
Speed in Vacuum	$c = 3 \times 10^8 \text{ m/s}$	Not applicable
Source	Accelerating charges	Vibrating particles
Can be Polarized?	Yes	Only transverse waves
Examples	Radio, Light, X-rays, etc.	Sound, Water waves, etc.



EXAM TIPS

- Always draw the EM wave diagram.
- Use key formulas: $c = \lambda \nu$, $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$, $I = \frac{E_0^2}{2\mu_0}$.
- Remember units: E (N/C), B (Tesla), S (W/m²).

24. MEMORY BOOSTER



- ★ Think: E for Electric (red), B for Magnetic (green).
- ★ Remember: $E \perp B \perp \text{Direction}$.
- ★ Speed in vacuum is constant (c).
- ★ More frequency $\Rightarrow$ Less wavelength.
- ★ Short λ , high ν , high energy (e.g., X-rays, Gamma rays).
- ★ Long λ , low ν , low energy (e.g., Radio waves).



★ DON'T FORGET

- EM waves = combination of oscillating E and B fields.
- They are the foundation of modern communication, technology and our understanding of the universe.



Understand Waves, Understand Nature $\rightarrow$ Keep Learning, Keep Exploring!

